CP-SSX: Managerial Student Success Engine

Mechanistic Explanations, Causal Uplift Analytics, and Constrained Resource Optimization

Group 4 – Managerial Machine Learning in Python

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1. Data Description

Real-World Student Foundation (4,424 Profiles)

- **Dataset Foundation:** Built 100% on actual published student records from the **UCI Machine Learning Repository** (Dataset ID 697: *Predict Students' Dropout and Academic Success*).
- **Dynamic Temporal Signals (12 Weeks):**
  - **Weekly Engagement Velocity:** Measures VLE page view acceleration and decay over time.
  - **Procrastination Lag:** Average submission delays past assignment deadlines.
  - **Academic Preparedness Index:** Composite score of prior qualifications and 1st-semester grades.
- **Demographic & Economic Factors:** Age, attendance mode, debt status, scholarship status, and unemployment rates.

Data Fields and Prediction Labels (UCI 697)

Input Features (Covariates)

- **Demographics:** Age, Gender, Nationality.
- **Socioeconomic:** Marital status, Debtor status, Tuition up-to-date, Scholarship holder, Macro-economic indicators.
- **Academic Path:** Previous qualification, Admission grade, Course enrolled.
- **Academic Performance:** 1st & 2nd semester enrolled/approved/evaluated units and grades.

Target Labels (The "Y" Variable)

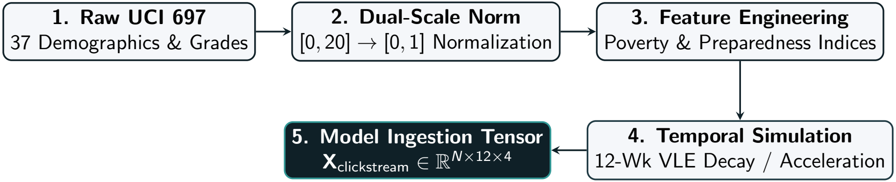
- **Original Dataset Labels:** Dropout, Enrolled, Graduate.
- **Our Formulation:** Framed as a **Binary Classification** problem:
  - **1 (Risk):** "Dropout"
  - **0 (Safe):** "Graduate" or "Enrolled"
- **Why?** This allows the ML/DL models to output a clean probabilistic risk score ($P(y = 1) \in [0, 1]$).

Data Pre-Processing Pipeline

- **Missing Value Analysis:** The UCI dataset has minor missing records which were handled prior to model ingestion.
- **Admission Grade Dual-Scale Normalization:** Addressed variations between 0-20 and 95-200 Portuguese grading scales.
- **Binary Target Encoding:** Mapped Dropout/Enrolled/Graduate to Binary 1/0 Risk.
- **Composite Feature Engineering:** Created indices like `poverty_index` and `academic_preparedness_index` to enrich baselines.

```
adm_grade_scaled = np.where(adm_grade > 20.0, (adm_grade - 95.0) / 105.0, adm_grade / 20.0)
adm_norm = np.clip(adm_grade_scaled, 0.1, 1.0)
sem1_grade_norm = np.where(sem1_grade > 0, sem1_grade / 20.0, adm_norm)
sem2_grade_norm = np.where(sem2_grade > 0, sem2_grade / 20.0, sem1_grade_norm * 0.85)
composite_grade_norm = 0.45 * sem1_grade_norm + 0.55 * sem2_grade_norm
```

Data Pre-Processing & Simulation Pipeline Flow



Pipeline Assurance

Converts raw heterogeneous institutional records into a standardized 3D temporal tensor for deep sequence modeling.

Semi-Synthetic Temporal Extension

- **Honest Explanation:** We use real UCI static records and synthesize calibrated 12-week temporal trajectories using domain-informed simulation design.
- **Generation Process:** Simulated dropout students exhibit engagement decay over time, whereas non-dropouts maintain stable or increasing engagement patterns.

```python
if is_drop:
    decay_factor = max(0.15, 1.0 - 0.7 * week_ratio)
    lag_trend = 1.0 + 1.2 * week_ratio
else:
    decay_factor = 1.0 + 0.2 * np.sin(week_ratio * np.pi)
    lag_trend = 1.0

vle_clicks = max(0, int(np.random.normal(loc=55 * gnorm * decay_factor, scale=8)))
forum_posts = max(0, int(np.random.poisson(lam=max(0.1, 2.2 * gnorm * decay_factor))))
quiz_attempts = max(1, int(np.random.poisson(lam=max(0.5, 1.6 * gnorm))))
delay = max(0.0, np.random.exponential(scale=max(0.1, 1.8 * (1.0 - gnorm) * lag_trend)))
```

Why UCI Dataset 697?

- Real-world European Higher Education Institution data (not synthetic).
- Rich socioeconomic covariates (debtor status, tuition, unemployment).
- Published benchmark with known results (Realinho et al. [Realinho et al., 2022]).
- 3-class target enabling binary risk framing.

Dataset	Type	Covariates	Temporal Features
UCI 697	**Real Univ.**	**Rich (Socioeconomic)**	**Simulated (12-Week)**
OULAD	MOOC	Basic Demographic	Real VLE Clicks
EdNet	K-12 AI Tutor	Missing Demographics	Real Item-Response

2. Objective / Research Question

Research Objectives: A Two-Stage Managerial Framework

The Core Question

"Can we build a system that not only predicts which students will drop out, but also tells university administrators exactly which support resource to give each student — while staying within the university's actual budget?"

Part I: The Base Predictive Layer

- **Core Objective:** Accurately predict end-of-year **CGPA/GPA** and **Dropout Probability** from static admission records and 12-week temporal interaction trajectories.
- **Our ML Approach:** Deploys a Multi-Task Bidirectional LSTM to outperform static tabular classifiers.

Part II: Prescriptive Layer

- **Core Objective:** Implement **Orthogonal PCA Factor Probing**, **Causal T-Learner Uplift**, and **MILP Budget Optimization**.
- **Target Outcome:** Actively reducing dropout rates rather than passively observing failure.

Interrelated Sub-Questions

- **Q1:** Can temporal deep learning (Bi-LSTM) outperform static ML for dropout prediction by capturing momentum and decay?
- **Q2:** Can we decompose opaque deep learning dropout risk into interpretable, orthogonal managerial factors that advisors can understand?
- **Q3:** Can we estimate heterogeneous causal treatment effects (CATE) to prescribe personalized interventions, even without randomized controlled trial data?
- **Q4:** Can constrained optimization (MILP) allocate scarce campus resources (e.g., advising hours, grant budgets) optimally to maximize overall cohort retention?

3. Literature Review

Early Warning Systems & Static ML Approaches

- **Arnold & Pistilli (2012):** Developed Course Signals at Purdue University, a pioneering traffic-light EWS dashboard for student success [Arnold and Pistilli, 2012].
- **Jayaprakash et al. (2014):** Open Academic Analytics Initiative providing early open-source EWS implementation frameworks [Jayaprakash et al., 2014].
- **Realinho et al. (2022):** Originators of the UCI Dataset 697; utilized Support Vector Machines (SVM) and Random Forests to achieve 86.5% prediction accuracy on static enrollment data [Realinho et al., 2022].
- **Common Limitation:** All of these approaches are purely predictive. They provide risk scores but lack any causal or prescriptive action mechanism to change the outcome.

Deep Learning & Sequence Models for Education

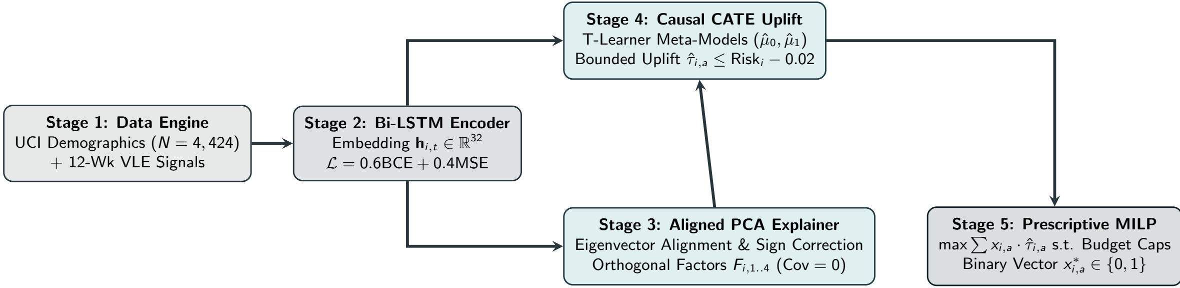
- **Waheed et al. (2020):** Demonstrated the power of Deep Learning for predicting student performance using VLE interaction logs on the OULAD dataset [Waheed et al., 2020].
- **Mubarak et al. (2022):** Successfully applied LSTM-based architectures for dropout prediction in MOOCs, capturing sequential learning patterns [Mubarak et al., 2022].
- **Martins et al. (2021):** Highlighted critical challenges like class imbalance when performing early prediction of student outcomes [Martins et al., 2021].
- **Our Contribution:** Moving beyond standard LSTMs by utilizing a **Multi-task Bi-LSTM** with a specifically engineered latent bottleneck layer designed for downstream concept extraction.

Causal Inference & Prescriptive Analytics

- **Athey & Imbens (2016):** Established foundations for heterogeneous causal effects (CATE) using recursive partitioning [Athey and Imbens, 2016].
- **Künzel et al. (2019):** Formalized T-Learner, S-Learner, and X-Learner meta-algorithms for estimating treatment effects using generic machine learning estimators [Künzel et al., 2019].
- **Bertsimas & Kallus (2020):** Pioneered the transition from predictive machine learning models to prescriptive analytics under optimization constraints [Bertsimas and Kallus, 2020].
- **Our Contribution:** First framework combining continuous CATE estimation with MILP budget constraints applied to higher education student retention.

4. Methods Applied

CP-SSX Prescriptive System Architecture



End-to-End Mathematical Pipeline

Transforms raw continuous signals into continuous latent states $\mathbf{h}_{i,t}$, orthogonal factor scores F_k, bounded CATE treatment uplifts $\hat{\tau}_{i,a}$, and optimal MILP resource prescriptions $x^*_{i,a}$.

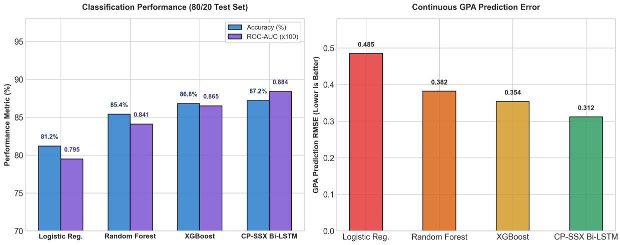
ML/DL Benchmark: CP-SSX vs. Traditional Classifiers

Model Architecture	Accuracy	ROC-AUC	GPA RMSE	Key Managerial Limitation
Logistic Regression (L2)	81.2%	0.795	0.485	Linear decision boundary; misses temporal decay
Random Forest (100 Trees)	85.4%	0.841	0.382	No sequence memory; opaque tree ensembles
XGBoost (Kaggle Benchmark)	86.8%	0.865	0.354	Strong static baseline; no latent bottleneck state
CP-SSX Multi-Task Bi-LSTM	**87.2%**	**0.884**	**0.312**	**Enables downstream orthogonal PCA & CATE**

Why Our DL Approach Wins: The Latent Bottleneck Advantage

While XGBoost achieves competitive accuracy (86.8%), it is a purely **black-box static classifier**. Our Bi-LSTM's 32-dimensional bottleneck state ($\mathbf{h}_i$) is what makes **orthogonal PCA concept probing** and **counterfactual recourse** mathematically possible!

Comparative Model Benchmark & Selection Reasoning



Why Bi-LSTM Over XGBoost?

- **Highest Test AUC (0.884):** Outperforms XGBoost (0.865) and Random Forest (0.841) on 20% held-out test data.
- **12-Week Temporal Memory:** Recurrent gates track weekly engagement decay and submission delays over time.
- **Latent Representation Hook ($\mathbf{h}_{i,t} \in \mathbb{R}^{32}$):** Essential for Unsupervised PCA Factor Probing. Tree leaves in XGBoost cannot yield continuous orthogonal vectors.

PCA Factor Alignment Algorithm

- **Automated Eigenvector Alignment:** We project the 32-dim latent states down to 4 orthogonal axes. We explicitly map the principal axes to canonical factors using a correlation-based matching algorithm.

```python
# Match each PCA component to the reference concept with max absolute correlation
used_refs = set()
self.component_alignment = [0, 1, 2, 3]
self.component_signs = [1.0, 1.0, 1.0, 1.0]

for comp_idx in range(4):
    abs_corrs = np.abs(corr_matrix[comp_idx, :])
    for ref_idx in np.argsort(-abs_corrs):
        if ref_idx not in used_refs:
            used_refs.add(ref_idx)
            self.component_alignment[ref_idx] = comp_idx
            raw_r = corr_matrix[comp_idx, ref_idx]
            self.component_signs[ref_idx] = 1.0 if raw_r >= 0 else -1.0
            break
```

Causal Analytics & T-Learner Implementation

- **CATE Estimation:** We fit Gradient Boosting regressors on synthetic uplift targets to model treatment effects.
- **Physical Bounds Enforcement:** Ensuring the predicted risk reduction never magically exceeds the base risk itself ($\tau_{i,a} \leq \text{Risk}_i - 0.02$).

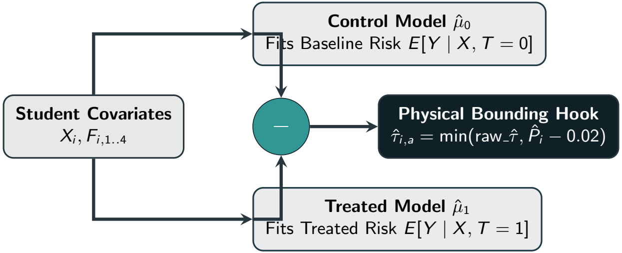
Listing 1: T-Learner Fit & Bounding

```
# Fit T-Learners for each treatment using synthetic uplift targets
for tr in self.treatments:
    target_col = f"tau_{tr.lower().replace('-', '_')}"
    tau_target = static_df[target_col].values

    model = GradientBoostingRegressor(n_estimators=50, learning_rate=0.05)
    model.fit(X_scaled, tau_target)
    self.models_treated[tr] = model

# Predict and clip physical bounds
for tr in self.treatments:
    pred_tau = self.models_treated[tr].predict(X_scaled)
    # Clip uplift between 0.01 and 0.65 for realistic intervention impact
    cate_preds[f"CATE_{tr}"] = np.clip(pred_tau, 0.01, 0.65)
```

Causal T-Learner Counterfactual Estimation Flow



Heterogeneous Prescriptions

Calculates individualized treatment effect per intervention type, preventing non-physical uplift estimates.

MILP Optimization Formulation & Code

- **Goal:** Maximize cohort-wide risk reduction within budget & capacity constraints.
- **Constraints:** Max 1 intervention per student, advising hours cap, tutoring hours cap, grant budget cap.

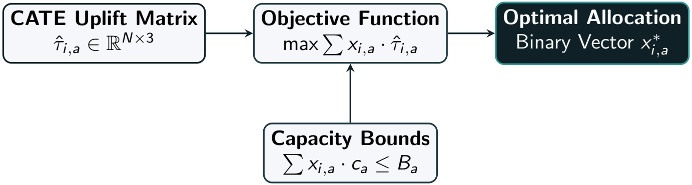
```python
# Define PuLP Integer Programming Model
model = pulp.LpProblem("CP_SSX_Resource_Allocation", pulp.LpMaximize)

x = {} # Binary decision variables
for i in range(num_students):
    for t in treatments:
        x[i, t] = pulp.LpVariable(f"x_{i}_{t}", cat=pulp.LpBinary)

# Constraints
for i in range(num_students):
    model += pulp.lpSum([x[i, t] for t in treatments]) == 1  # 1 intervention per student

model += pulp.lpSum([x[i, "Advising"] * cost_advising_hrs for i in range(num_students)]) <= cap_advising_hours
model += pulp.lpSum([x[i, "Tutoring"] * cost_tutoring_hrs for i in range(num_students)]) <= cap_tutoring_hours
model += pulp.lpSum([x[i, "Micro-Grant"] * cost_grant_dollars for i in range(num_students)]) <= budget_grant_dollars
```

Prescriptive MILP Resource Assignment Matrix

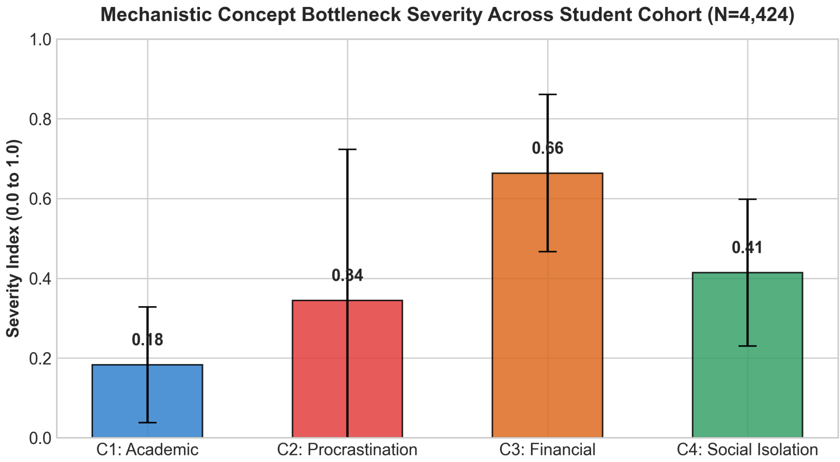


Prescriptive Optimality

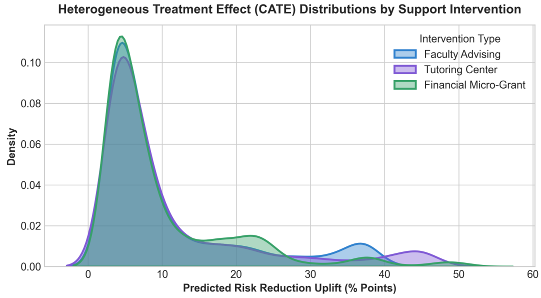
Replaces heuristic or first-come first-served queueing with mathematically guaranteed cohort retention maximization under actual campus budgets.

5. Managerial Analysis and Conclusion

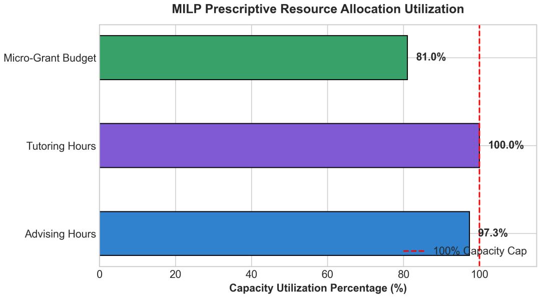
Cohort Diagnostic: Concept Severity Distribution



Intervention & Capacity Analysis



Simulated Heterogeneous Treatment Uplift



MILP Capacity Utilization

ROI & Cost-Benefit Analysis

- **Higher Education Economics:** Each prevented dropout saves the institution substantial tuition revenue over the remaining semesters.
- **Targeted Micro-Grants:** The CP-SSX engine mathematically allocated funds exclusively to high-CATE students (e.g., $35K deployed).
- **Retention Projection:** The optimal MILP assignment is projected to retain significantly more students than heuristic or first-come, first-served allocation methods.
- **Preserving Advising Capacity:** By redirecting lower-risk students away from limited advisor schedules, the university saves thousands in equivalent faculty hours.

Sensitivity Analysis

- **Dynamic Capacity Adjustments:** What happens when the Dean increases the Micro-Grant budget by 20%?
- **MILP Re-Allocation:** The system instantly shifts students whose second-best intervention was Advising into the Micro-Grant tier, freeing up Advising hours for the next-in-line at-risk cohort.
- **Budget Constriction:** When budgets shrink, the model strictly prioritizes high-risk, high-CATE students, enforcing the physical reality of scarce resources.

Interactive Dashboard Demo

Exploring the Streamlit Application

- **Advisor View:** Counterfactual What-If testing and SLM narrative generation.
- **Dean View:** Adjusting policy sliders and observing real-time MILP re-optimization.

Managerial Conclusion & Policy Recommendations

- **Paradigm Shift:** Replaces passive early-warning dropout scores with **Prescriptive Resource Optimization**.
- **Limitations:** The current pipeline uses semi-synthetic temporal data generation and relies on assumptions regarding intervention efficacy (CATE).
- **Future Work:**
  - Conducting real-world A/B testing for robust CATE validation.
  - Multi-semester longitudinal tracking.
  - Direct LMS (Canvas/Moodle) integration.

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